

Bilingual Infant Noise (BIN) Study

Word Recognition & Habituation — Results Walkthrough

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Word Recognition Task

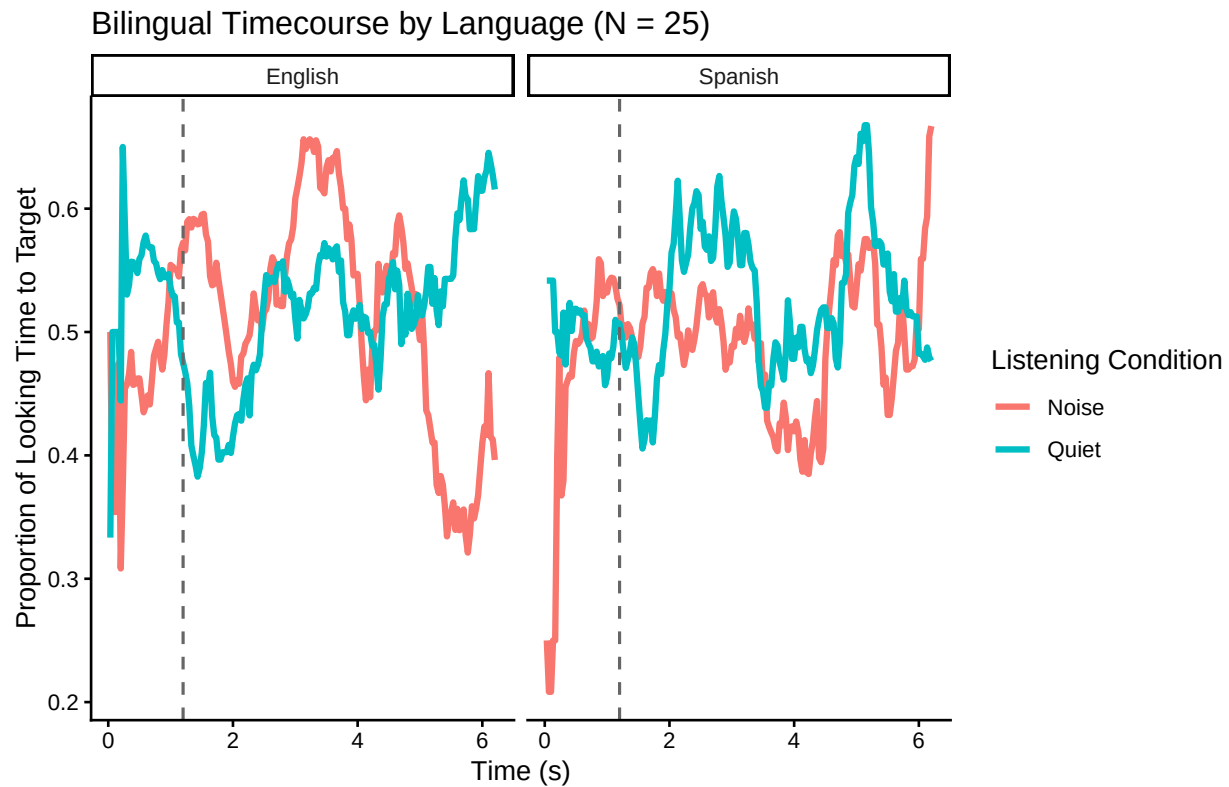
Bilingual Timecourse

There are 16 trial types for bilinguals:

- quiet (8): stroller, hand, car, foot, perro, nariz, leche, ojo
- noise (8): hand, foot, stroller, car, perro, nariz, ojo, leche

Object pairs: ojo/perro, car/foot, stroller/hand, nariz/leche.

Includes bilinguals over 12 months old ($N = 25$, updated 7/20/26). The dashed line at 1.2 s marks target word onset.



Bilingual Accuracy (baseline-corrected)

Looking-time data for a 4 s window (2–6 s); baseline-corrected using the 0–1.2 s window.

Baseline survival. The baseline window is only 36 frames, so requiring 15 (= 0.5 s) is a high bar. How many trials survive before we trust the corrected numbers:

Bilingual trials with valid TEST window: 354 | with valid test AND baseline: 334 (94% retained)

Inclusion rule: keep participants with ≥ 4 valid corrected Quiet *and* ≥ 4 valid corrected Noise trials.

Kept 23 of 25 participants

SES join. IDs are normalized on both sides before joining. The check below should return an empty tibble — if it lists anyone, those looking-data IDs still don't match a row in `predictors.csv`.

```
# A tibble: 9 x 1
  Participant
  <chr>
1 B11ARF
2 B12KRM
```

3 B14I
 4 B17YT
 5 B25ELM
 6 B29NRP
 7 B31ZR
 8 B35VC
 9 B41RRG

Participant-level means

```
# A tibble: 4 x 5
  quiet_noise language mean_corr      se      n
  <chr>         <chr>      <dbl> <dbl> <int>
1 Noise        English    0.0224 0.0409   23
2 Noise        Spanish    0.00968 0.0340   23
3 Quiet        English   -0.0628 0.0522   23
4 Quiet        Spanish    0.0192 0.0330   23
```

Tests against 0 (null = no shift from baseline)

One Sample t-test

```
data: p_overall$acc
t = -0.33378, df = 22, p-value = 0.7417
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 -0.03811307  0.02754560
sample estimates:
mean of x
-0.005283737
```

Paired t-test

```
data: by_cond_wide$Quiet and by_cond_wide$Noise
t = -0.63306, df = 22, p-value = 0.5332
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -0.12696336  0.06757834
sample estimates:
mean difference
-0.02969251
```

Split by language

```
# A tibble: 4 x 10
  language quiet_noise    N   Mean   SE CI_low CI_high    t    df    p
  <chr>      <chr>      <int> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 English  Noise           23  0.022  0.041 -0.062  0.107  0.547   22 0.590
2 English  Quiet           23 -0.063  0.052 -0.171  0.046 -1.20   22 0.242
3 Spanish  Noise           23  0.01   0.034 -0.061  0.08   0.285   22 0.778
4 Spanish  Quiet           23  0.019  0.033 -0.049  0.088  0.582   22 0.566
```

Mixed-effects model, with vs. without SES

Both models are fit on the SES-complete subset so the comparison is apples-to-apples. `anova()` tests whether adding parent education improves fit.

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [lmerModLmerTest]
Formula: corrected_acc ~ quiet_noise * language + parent_ed_z + (1 | Participant)
Data: b_ses_complete
```

REML criterion at convergence: 224.2

Scaled residuals:

Min	1Q	Median	3Q	Max
-2.42288	-0.62931	-0.00179	0.58580	2.55522

Random effects:

Groups	Name	Variance	Std.Dev.
Participant	(Intercept)	0.0000	0.0000
Residual		0.1793	0.4234

Number of obs: 188, groups: Participant, 14

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	-0.008461	0.079289	183.000000	-0.107	0.915
quiet_noiseQuiet	-0.086197	0.087364	183.000000	-0.987	0.325
languageSpanish	0.023752	0.087406	183.000000	0.272	0.786
parent_ed_z	-0.010216	0.056330	183.000000	-0.181	0.856
quiet_noiseQuiet:languageSpanish	0.095556	0.123547	183.000000	0.773	0.440

Correlation of Fixed Effects:

```
(Intr) qt_nsQ lnggSp prnt__
quiet_nosQt -0.545
langugSpnsh -0.518  0.489
parent_ed_z  0.637 -0.010  0.032
qt_nsQt:lnS  0.378 -0.707 -0.707 -0.005
optimizer (nloptwrap) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')
```

Data: b_ses_complete

Models:

m0: corrected_acc ~ quiet_noise * language + (1 | Participant)

m1: corrected_acc ~ quiet_noise * language + parent_ed_z + (1 | Participant)

	npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
m0	6	217.34	236.75	-102.67	205.34			
m1	7	219.30	241.96	-102.65	205.30	0.0338	1	0.8542

N participants in SES models: 14

For reference, the same model on the **full** bilingual sample (no SES filter):

Linear mixed model fit by REML. t-tests use Satterthwaite's method [lmerModLmerTest]

Formula: corrected_acc ~ quiet_noise * language + (1 | Participant)

Data: bcorr_clean

REML criterion at convergence: 385.1

Scaled residuals:

Min	1Q	Median	3Q	Max
-2.36347	-0.62011	-0.04261	0.62231	2.44817

Random effects:

Groups	Name	Variance	Std.Dev.
Participant	(Intercept)	0.0000	0.0000
Residual		0.1888	0.4345

Number of obs: 318, groups: Participant, 23

Fixed effects:

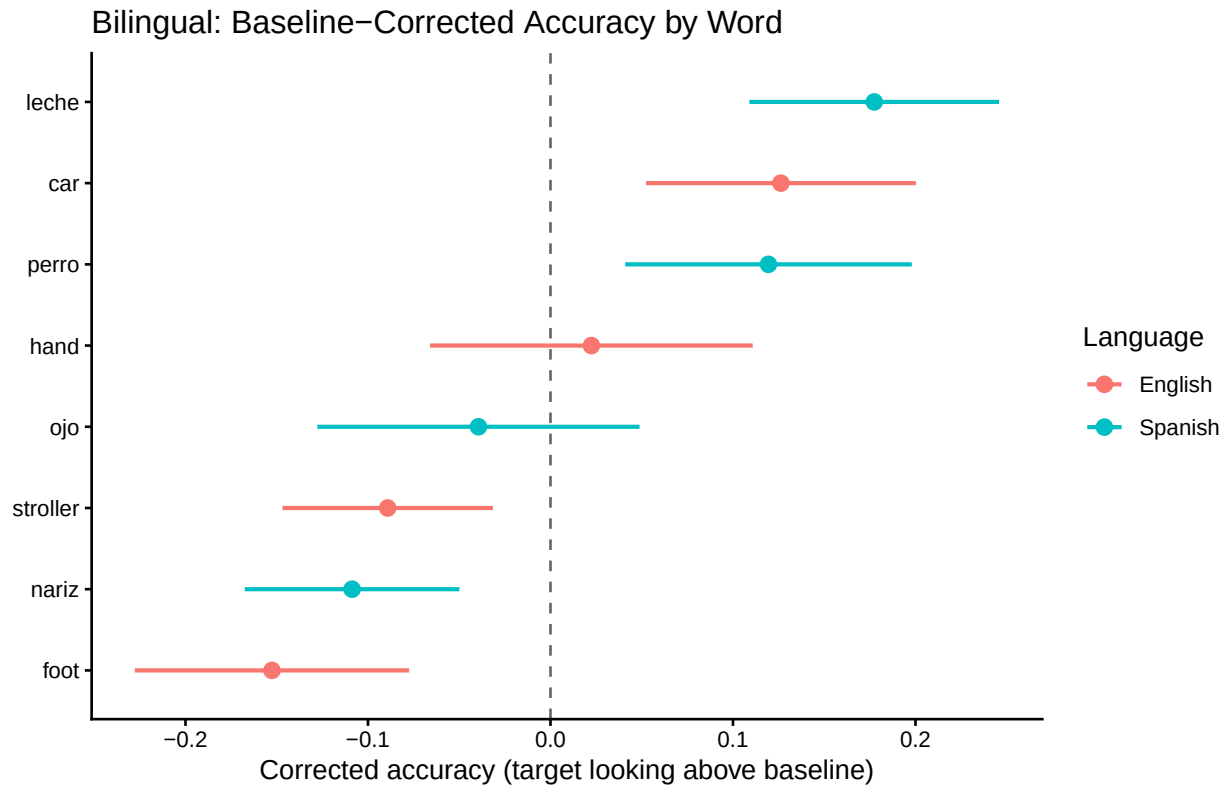
	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	0.018514	0.048576	314.000000	0.381	0.703
quiet_noiseQuiet	-0.082188	0.068697	314.000000	-1.196	0.232
languageSpanish	-0.009606	0.068697	314.000000	-0.140	0.889
quiet_noiseQuiet:languageSpanish	0.100156	0.097463	314.000000	1.028	0.305

Correlation of Fixed Effects:

	(Intr)	qt_nsQ	lnggSp
quiet_nosQt	-0.707		
langugSpnsh	-0.707	0.500	
qt_nsQt:lnS	0.498	-0.705	-0.705

optimizer (nloptwrap) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')

Accuracy by object



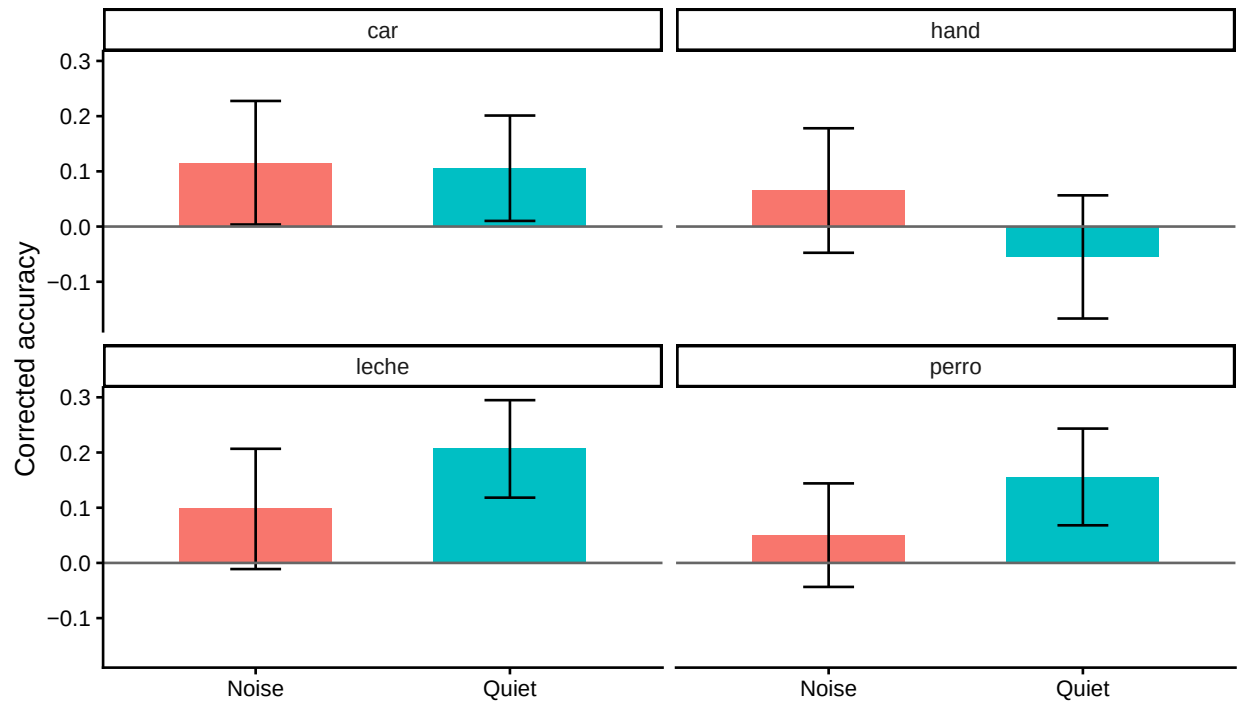
Recognized words (exploratory)

Words selected post-hoc for being above 0.

```
# A tibble: 8 x 5
  word quiet_noise mean se n
  <chr> <chr>      <dbl> <dbl> <int>
1 car Noise      0.116 0.112 19
2 car Quiet      0.106 0.0954 21
3 hand Noise      0.0653 0.113 20
4 hand Quiet     -0.0550 0.112 19
5 leche Noise      0.0978 0.109 18
6 leche Quiet      0.207 0.0883 22
7 perro Noise      0.0503 0.0938 22
8 perro Quiet      0.156 0.0876 18
```

Bilingual: Quiet vs. Noise for Recognized Words

Exploratory – words selected post-hoc for being above 0



Paired t-test

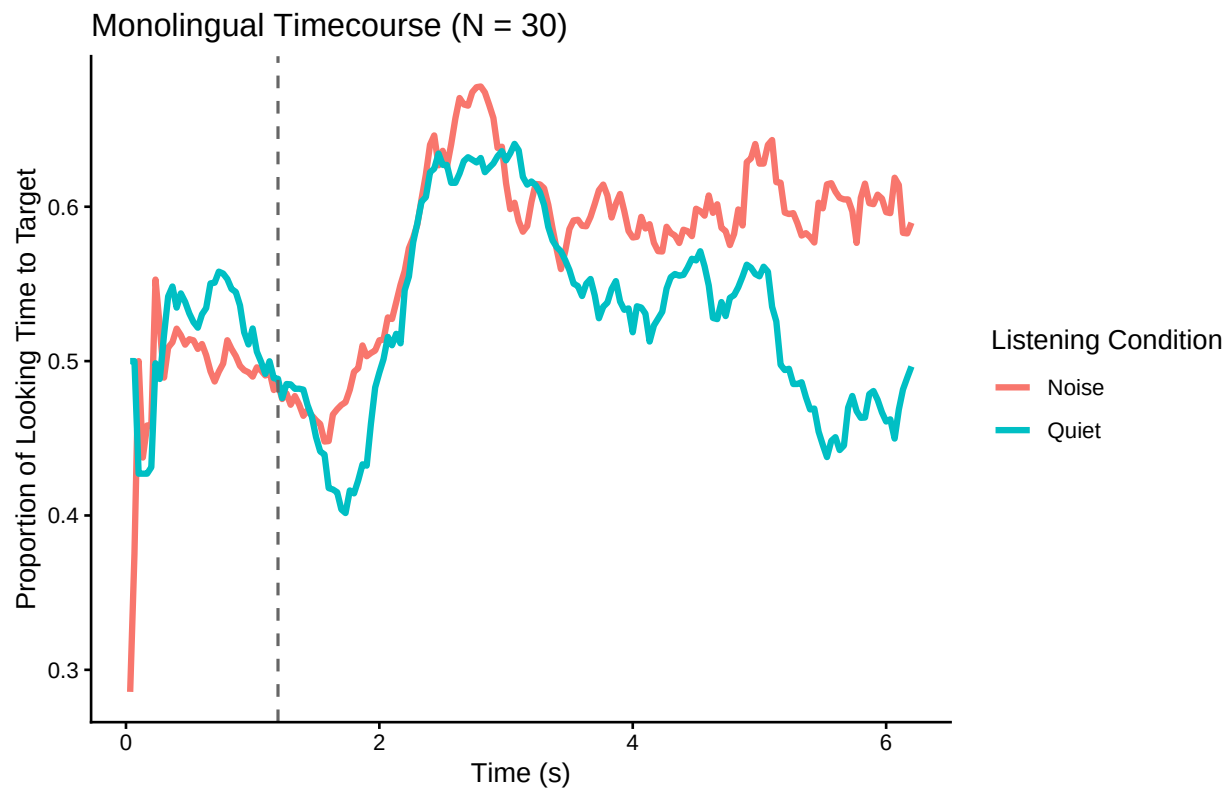
```
data: b_good_wide$Quiet and b_good_wide$Noise
t = 0.76553, df = 22, p-value = 0.4521
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -0.09750527  0.21160784
sample estimates:
mean difference
 0.05705129
```

Monolingual Timecourse

16 trial types for monolinguals:

- quiet (8): stroller, hand, car, foot, eye, nose, dog, milk
- noise (8): hand, foot, stroller, car, eye, nose, milk, dog

Object pairs: eye/dog, car/foot, stroller/hand, milk/nose. Includes monolinguals over 12 months old (N = 30, updated 7/20/26).



Monolingual Accuracy (baseline-corrected)

Baseline survival:

Monolingual trials with valid TEST window: 460 | with valid test AND baseline: 444 (97% retained)

Inclusion rule (same as bilinguals):

Kept 28 of 30 participants

```
# A tibble: 6 x 1
  Participant
  <chr>
1 M07CR
2 M17EW
3 M36EW
4 M37JD
5 M49AW
6 M50SS
```

Participant-level means

```
# A tibble: 2 x 4
  quiet_noise mean_corr      se      n
  <chr>         <dbl>   <dbl> <int>
1 Noise         0.0901 0.0283    28
2 Quiet         0.0172 0.0284    28
```

Tests against zero

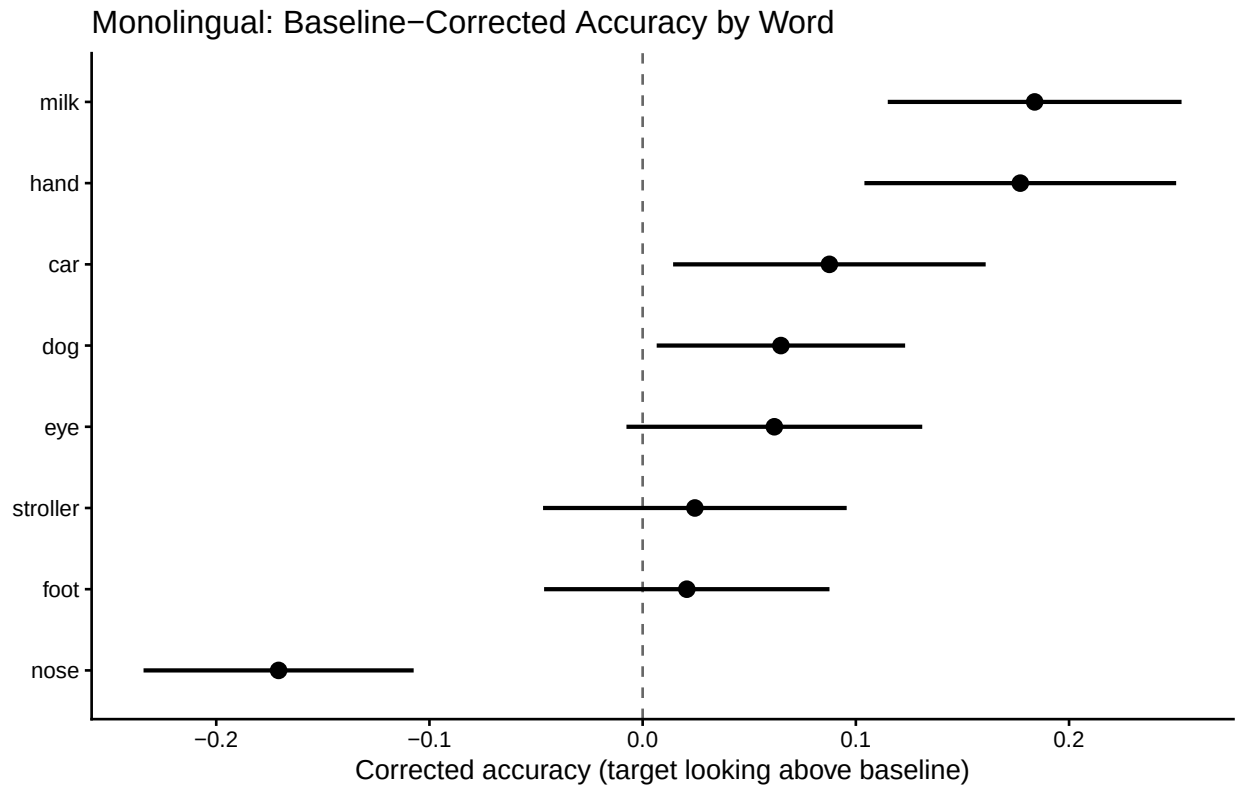
One Sample t-test

```
data: m_overall$acc
t = 2.26, df = 27, p-value = 0.03209
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 0.004953272 0.102577876
sample estimates:
mean of x
0.05376557
```

Paired t-test

```
data: m_by_cond_wide$Quiet and m_by_cond_wide$Noise
t = -2.2431, df = 27, p-value = 0.0333
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
-0.139593821 -0.006215308
sample estimates:
mean difference
-0.07290456
```

Accuracy by object

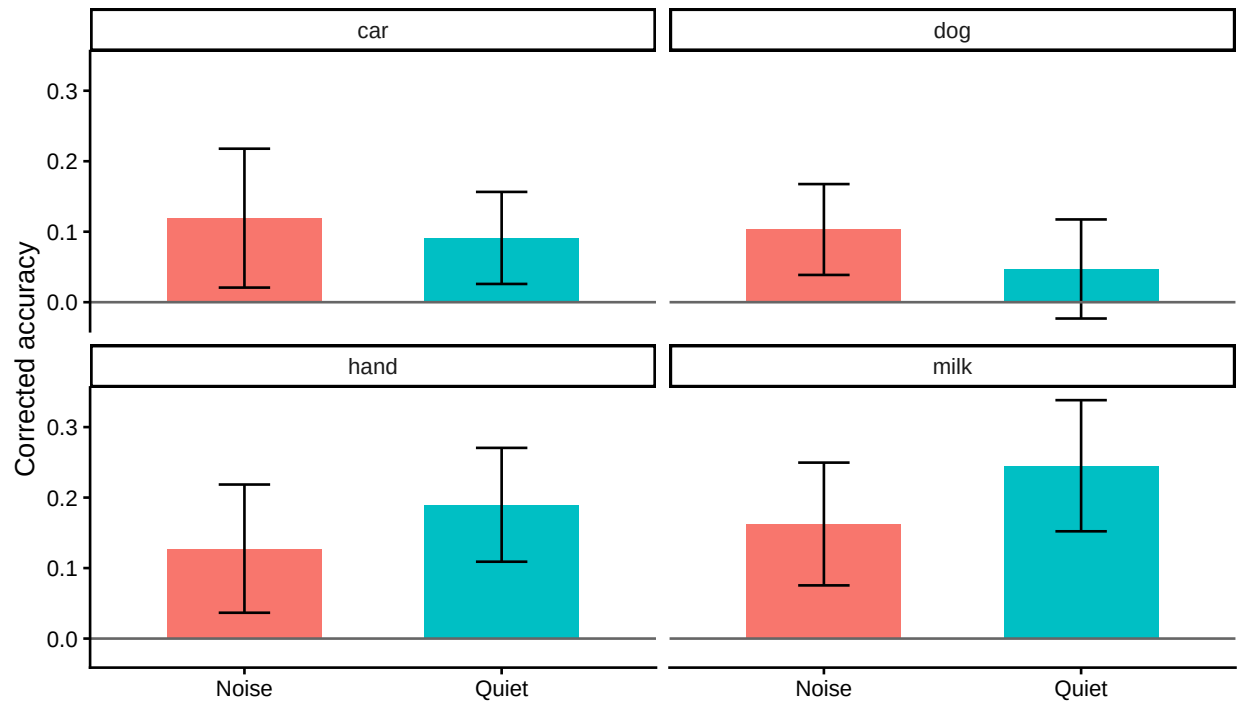


Recognized words (exploratory)

```
# A tibble: 8 x 5
  word quiet_noise mean se n
  <chr> <chr>      <dbl> <dbl> <int>
1 car   Noise      0.119 0.0985 25
2 car   Quiet      0.0912 0.0653 26
3 dog   Noise      0.103 0.0644 26
4 dog   Quiet      0.0471 0.0703 28
5 hand  Noise      0.128 0.0909 28
6 hand  Quiet      0.190 0.0807 25
7 milk  Noise      0.163 0.0871 24
8 milk  Quiet      0.245 0.0930 23
```

Monolingual: Quiet vs. Noise for Recognized Words

Exploratory – words selected post-hoc for being above 0



Paired t-test

```
data: m_good_wide$Quiet and m_good_wide$Noise
t = 0.025715, df = 27, p-value = 0.9797
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -0.1066480  0.1093551
sample estimates:
mean difference
 0.001353531
```

Mixed-effects model, with vs. without SES

Little is expected here: monolingual parent education has a restricted range (mostly 16–17 years).

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [lmerModLmerTest]
Formula: corrected_acc ~ quiet_noise + parent_ed_z + (1 | Participant)
Data: m_ses_complete
```

REML criterion at convergence: 398

Scaled residuals:

Min	1Q	Median	3Q	Max
-2.1836	-0.6785	-0.1120	0.7285	2.2819

Random effects:

Groups	Name	Variance	Std.Dev.
Participant	(Intercept)	0.004003	0.06327
	Residual	0.179599	0.42379

Number of obs: 339, groups: Participant, 22

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	0.10402	0.04051	41.19426	2.568	0.014 *
quiet_noiseQuiet	-0.05957	0.04606	318.22133	-1.293	0.197
parent_ed_z	-0.02354	0.03473	14.57278	-0.678	0.509

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	qt_nsQ
quiet_nosQt	-0.560	
parent_ed_z	-0.490	-0.014

Data: m_ses_complete

Models:

	npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
mm0: corrected_acc ~ quiet_noise + (1 Participant)	4	391.85	407.15	-191.93	383.85			
mm1: corrected_acc ~ quiet_noise + parent_ed_z + (1 Participant)	5	393.30	412.43	-191.65	383.30	0.5443	1	0.4606

N participants in SES models: 22

Linear mixed model fit by REML. t-tests use Satterthwaite's method [lmerModLmerTest]

Formula: corrected_acc ~ quiet_noise + (1 | Participant)

Data: mcorr_clean

REML criterion at convergence: 498.5

Scaled residuals:

Min	1Q	Median	3Q	Max
-2.4623	-0.6707	-0.1087	0.7284	2.3952

Random effects:

Groups	Name	Variance	Std.Dev.
Participant	(Intercept)	0.003469	0.0589
	Residual	0.180533	0.4249

Number of obs: 429, groups: Participant, 28

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	0.08453	0.03100	79.65918	2.727	0.00786 **
quiet_noiseQuiet	-0.06390	0.04105	403.73858	-1.557	0.12031

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

(Intr)
quiet_nosQt -0.655

Combined Analyses

Summary table (both groups vs. 0)

A tibble: 2 x 8

Group	N	Mean	CI_low	CI_high	t	df	p
<chr>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1 Monolingual	28	0.054	0.005	0.103	2.26	27	0.0321
2 Bilingual	23	-0.005	-0.038	0.028	-0.334	22	0.742

Matched-object model

Bilinguals heard leche/perro in Spanish; monolinguals heard milk/dog in English — same physical objects, collapsed to one label. The four shared objects are milk, dog, hand, car.

Rows: 372 | Participants: 51 | Objects: 4

A tibble: 2 x 2

group	n
<fct>	<int>
1 Monolingual	28
2 Bilingual	23

Linear mixed model fit by REML. t-tests use Satterthwaite's method [lmerModLmerTest]
Formula: corrected_acc ~ group * quiet_noise + (1 | Participant) + (1 | object)
Data: matched

REML criterion at convergence: 439.7

Scaled residuals:

Min	1Q	Median	3Q	Max
-2.6099	-0.6986	-0.1240	0.6717	2.1425

Random effects:

Groups	Name	Variance	Std.Dev.
Participant	(Intercept)	0.000e+00	0.000e+00
object	(Intercept)	4.077e-20	2.019e-10
Residual		1.841e-01	4.291e-01

Number of obs: 372, groups: Participant, 51; object, 4

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	0.129077	0.041678	368.000000	3.097	0.0021 **
groupBilingual	-0.022572	0.063551	368.000000	-0.355	0.7227
quiet_noiseNoise	-0.009144	0.058804	368.000000	-0.155	0.8765
groupBilingual:quiet_noiseNoise	-0.016735	0.089946	368.000000	-0.186	0.8525

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

(Intr) grpBln qt_nsN

groupBilngl	-0.656
quiet_nosNs	-0.709 0.465
grpBilngl:_N	0.463 -0.707 -0.654

optimizer (nloptwrap) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')

group	quiet_noise	emmean	SE	df	lower.CL	upper.CL
Monolingual	Quiet	0.1291	0.0418	29.9	0.04365	0.215
Bilingual	Quiet	0.1065	0.0480	48.4	0.00992	0.203
Monolingual	Noise	0.1199	0.0416	29.4	0.03487	0.205
Bilingual	Noise	0.0806	0.0483	49.6	-0.01651	0.178

Degrees-of-freedom method: kenward-roger

Confidence level used: 0.95

quiet_noise = Quiet:

contrast	estimate	SE	df	t.ratio	p.value
Monolingual - Bilingual	0.0226	0.0637	164	0.354	0.7237

quiet_noise = Noise:

contrast	estimate	SE	df	t.ratio	p.value
Monolingual - Bilingual	0.0393	0.0638	165	0.616	0.5385

Degrees-of-freedom method: kenward-roger

group = Monolingual:

contrast	estimate	SE	df	t.ratio	p.value
Quiet - Noise	0.00914	0.0589	324	0.155	0.8766

```
group = Bilingual:
  contrast      estimate      SE  df t.ratio p.value
Quiet - Noise  0.02588 0.0682 327   0.379  0.7046
```

Degrees-of-freedom method: kenward-roger

Matched-object model with SES

This is where group and SES are nearly confounded. Report both models and watch the `group` coefficient. The collinearity check and VIF quantify how separable the two are in this subset.

```
# A tibble: 2 x 6
  group      n mean_ed sd_ed  min  max
  <fct>    <int>   <dbl> <dbl> <dbl> <dbl>
1 Monolingual  22    16.8  1.98   13   21
2 Bilingual   14    12.9  1.56    9   15
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [lmerModLmerTest]
Formula: corrected_acc ~ group * quiet_noise + parent_ed_z + (1 | Participant) +
  (1 | object)
Data: matched_ses
```

REML criterion at convergence: 301.4

```
Scaled residuals:
    Min       1Q   Median       3Q      Max
-2.61170 -0.68250 -0.06589  0.67981  2.19559
```

```
Random effects:
 Groups      Name      Variance Std.Dev.
Participant (Intercept) 0.0000000 0.00000
object      (Intercept) 0.0004324 0.02079
Residual                    0.1737465 0.41683
Number of obs: 262, groups: Participant, 36; object, 4
```

```
Fixed effects:
              Estimate Std. Error      df t value Pr(>|t|)
(Intercept)    0.17819    0.05133  31.31718   3.472  0.00153 **
groupBilingual -0.13621    0.09363 254.73829  -1.455  0.14697
quiet_noiseNoise -0.03618    0.06395 254.22590  -0.566  0.57208
parent_ed_z    -0.05317    0.03595 254.80801  -1.479  0.14040
groupBilingual:quiet_noiseNoise  0.05810    0.10794 255.49535   0.538  0.59085
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Correlation of Fixed Effects:

```

      (Intr) grpBln qt_nsN prnt__
groupBln gl -0.675
quiet_nosNs -0.622  0.346
parent_ed_z -0.438  0.582  0.014
grpBln gl:_N  0.359 -0.566 -0.592  0.014
optimizer (nloptwrap) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')

```

Data: matched_ses

Models:

```

c0: corrected_acc ~ group * quiet_noise + (1 | Participant) + (1 | object)
c1: corrected_acc ~ group * quiet_noise + parent_ed_z + (1 | Participant) + (1 | object)
      npar      AIC      BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
c0      7 296.65 321.63 -141.33    282.65
c1      8 296.43 324.98 -140.21    280.43 2.2237  1    0.1359

```

group	quiet_noise	parent_ed_z	group:quiet_noise
3.010943	1.541562	2.047580	2.542794

Within-bilingual SES effect

Does the Quiet–Noise gap shrink or grow with SES? (quiet_noise:parent_ed_z is the term of interest.)

```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [lmerModLmerTest]
Formula: corrected_acc ~ quiet_noise * parent_ed_z + language + (1 | Participant)
Data: b_ses_complete

```

REML criterion at convergence: 223.4

Scaled residuals:

Min	1Q	Median	3Q	Max
-2.24360	-0.58600	0.04658	0.58882	2.44648

Random effects:

Groups	Name	Variance	Std.Dev.
Participant	(Intercept)	0.0000	0.0000
Residual		0.1783	0.4222

Number of obs: 188, groups: Participant, 14

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	0.03397	0.08970	183.00000	0.379	0.705
quiet_noiseQuiet	-0.16795	0.11945	183.00000	-1.406	0.161
parent_ed_z	0.06144	0.07962	183.00000	0.772	0.441
languageSpanish	0.07196	0.06166	183.00000	1.167	0.245

```
quiet_noiseQuiet:parent_ed_z  -0.14210    0.11226 183.00000  -1.266    0.207
```

Correlation of Fixed Effects:

```
(Intr) qt_nsQ prnt__ lnggSp
quiet_nosQt -0.674
parent_ed_z  0.807 -0.614
langugSpnsh -0.310 -0.016  0.033
qt_nsQt:p__ -0.578  0.857 -0.709 -0.005
optimizer (nloptwrap) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')
```

Habituation Task

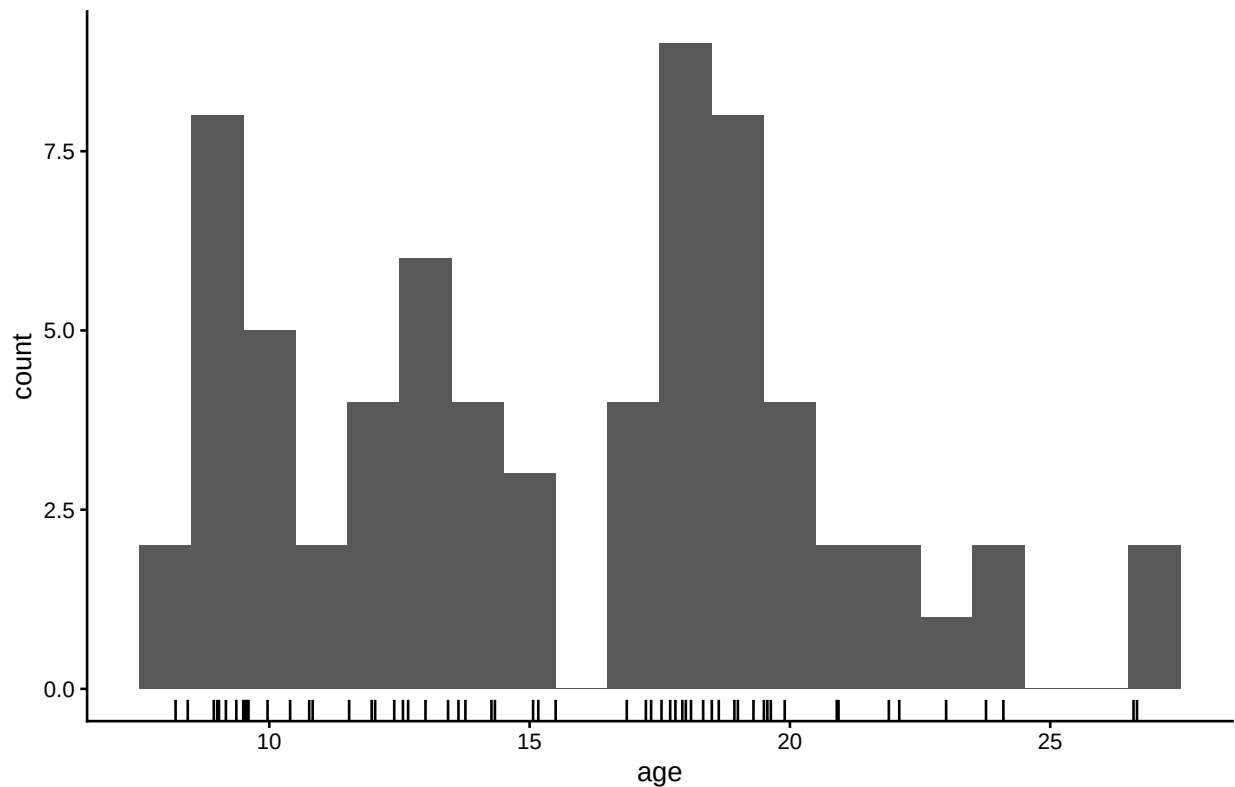
79 participants have habituation data (V1 and V2 points combined), all of whom habituated. We keep one point per participant, preferring V2 when present.

Missing Data

```
[1] "B04DR"      "B06JR"      "B10KG"      "B12KRM_V2" "B17YT"      "B22JR"
[7] "B23RF"      "B24EBR"     "B29NRP"     "B31ZR"      "B32NF"      "B36GBG"
[13] "M05ES"      "M08SA_V2"   "M24JB_V2"   "M37JD_V2"   "M43EW"      "M47JJ"
[19] "M48DZ"      "M49AW"      "M50SS"      "M51NS"
```

```
[1] "B17YT" "B23RF" "B29NRP" "B31ZR" "B36GBG" "M49AW" "M50SS" "M51NS"
```

Age Distribution



Regressions

Baseline grid (no SES)

A tibble: 15 x 8

	outcome	predictor	sample	term	estimate	std.error	statistic	p.value
	<chr>	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	nov_score	tv	bilingual	tv	7.18e-3	3.52e-3	2.04	0.0570
2	hab_trials	age * language	full	age:la~	-1.16e-1	6.84e-2	-1.70	0.0945
3	hab_trials	age * language	full	age	6.96e-2	4.76e-2	1.46	0.149
4	nov1	age * language	full	age	1.73e+2	1.35e+2	1.28	0.206
5	hab_trials	tv	bilingual	tv	-4.04e-2	3.14e-2	-1.29	0.215
6	hab_trials	age * language	full	langua~	1.32e+0	1.11e+0	1.19	0.238
7	nov1	noise_sum	monolingual	noise_~	1.18e+2	1.10e+2	1.07	0.293
8	nov1	age * language	full	langua~	-3.08e+3	3.16e+3	-0.974	0.334
9	nov_score	age * language	full	langua~	-1.82e-1	2.00e-1	-0.909	0.367
10	nov1	tv	bilingual	tv	6.85e+1	7.57e+1	0.906	0.378
11	nov_score	age * language	full	age:la~	8.25e-3	1.23e-2	0.672	0.504
12	nov1	age * language	full	age:la~	1.20e+2	1.94e+2	0.617	0.540
13	nov_score	age * language	full	age	2.03e-3	8.56e-3	0.238	0.813
14	hab_trials	noise_sum	monolingual	noise_~	8.13e-3	3.59e-2	0.226	0.823

15	nov_score	noise_sum	monolingual noise_~	1.45e-3	7.24e-3	0.201	0.843
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SES-augmented grid

Every predictor crossed with every sample (dropping language terms in single-group samples).
n_used tracks how SES missingness shrinks each model.

A tibble: 84 x 9

	outcome	predictor	sample	n_used	term	estimate	std.error	statistic	p.value
	<chr>	<chr>	<chr>	<int>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	nov_score	tv + parent_e~	full	46	tv	8.74e-3	0.00428	2.04	0.0472
2	nov_score	tv	bilin~	19	tv	7.18e-3	0.00352	2.04	0.0570
3	nov_score	tv	full	46	tv	7.42e-3	0.00393	1.89	0.0660
4	nov1	tv + parent_e~	bilin~	19	pare~	1.74e+3	880.	1.97	0.0662
5	nov_score	tv + parent_e~	bilin~	19	tv	7.15e-3	0.00363	1.97	0.0663
6	nov1	tv + parent_e~	full	46	tv	1.31e+2	70.6	1.85	0.0714
7	nov1	noise_sum + p~	full	46	nois~	1.00e+2	55.9	1.80	0.0795
8	hab_trials	age * language	full	68	age::~	-1.16e-1	0.0684	-1.70	0.0945
9	nov_score	noise_sum + p~	monol~	27	pare~	1.18e-1	0.0689	1.71	0.0995
10	nov1	noise_sum + p~	bilin~	19	pare~	1.60e+3	925.	1.73	0.102

i 74 more rows

Confirmatory model

Pre-specified after the exploratory grid: full sample, SES entered as a covariate alongside the predictor of interest (tv). Report N explicitly.

Call:

```
lm(formula = nov_score ~ tv + age + language + parent_ed_z, data = confirm_df)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.51000	-0.14819	0.01607	0.13935	0.37164

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.564407	0.133929	4.214	0.000134 ***
tv	0.006467	0.004629	1.397	0.169933
age	0.008908	0.007720	1.154	0.255216
language	-0.129110	0.102574	-1.259	0.215259
parent_ed_z	0.059925	0.042476	1.411	0.165843

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.227 on 41 degrees of freedom

Multiple R-squared: 0.1382, Adjusted R-squared: 0.05418
F-statistic: 1.644 on 4 and 41 DF, p-value: 0.1815

A tibble: 5 x 7

term	estimate	std.error	statistic	p.value	conf.low	conf.high
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1 (Intercept)	0.564	0.134	4.21	0.000134	0.294	0.835
2 tv	0.00647	0.00463	1.40	0.170	-0.00288	0.0158
3 age	0.00891	0.00772	1.15	0.255	-0.00668	0.0245
4 languagemonolingual	-0.129	0.103	-1.26	0.215	-0.336	0.0780
5 parent_ed_z	0.0599	0.0425	1.41	0.166	-0.0259	0.146

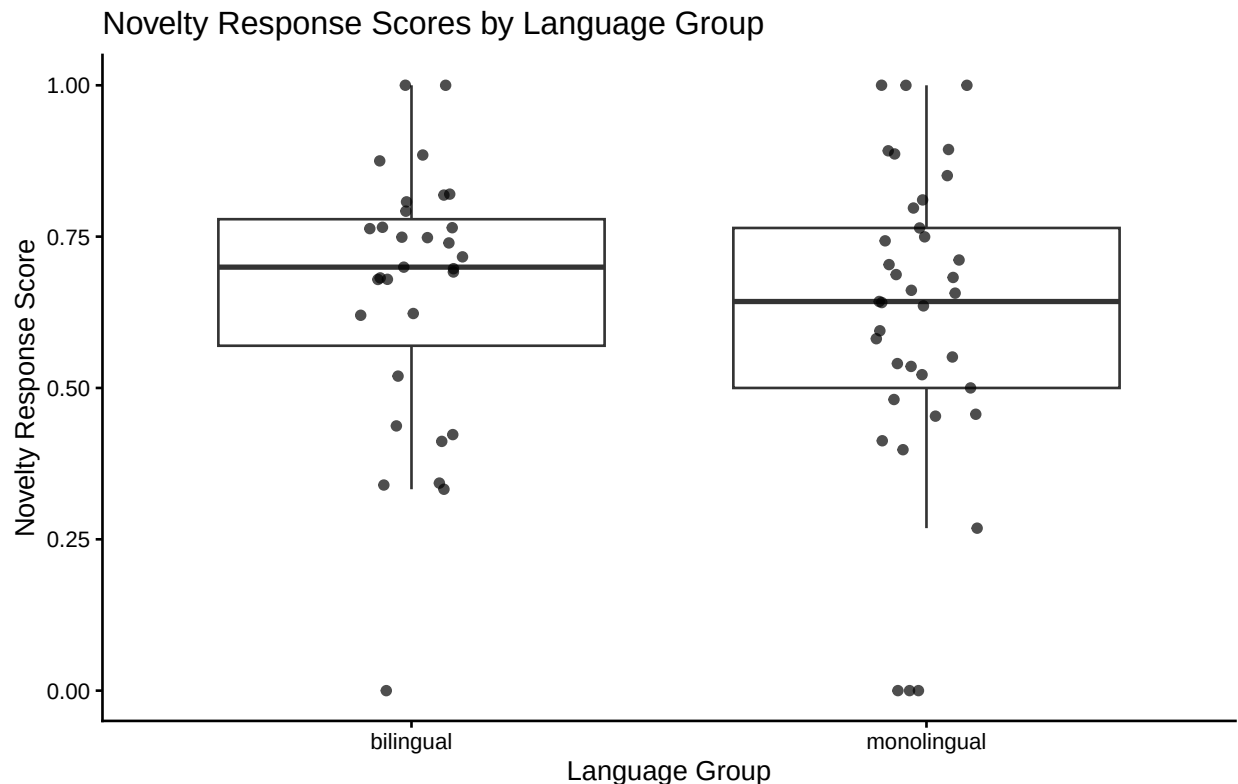
A tibble: 1 x 12

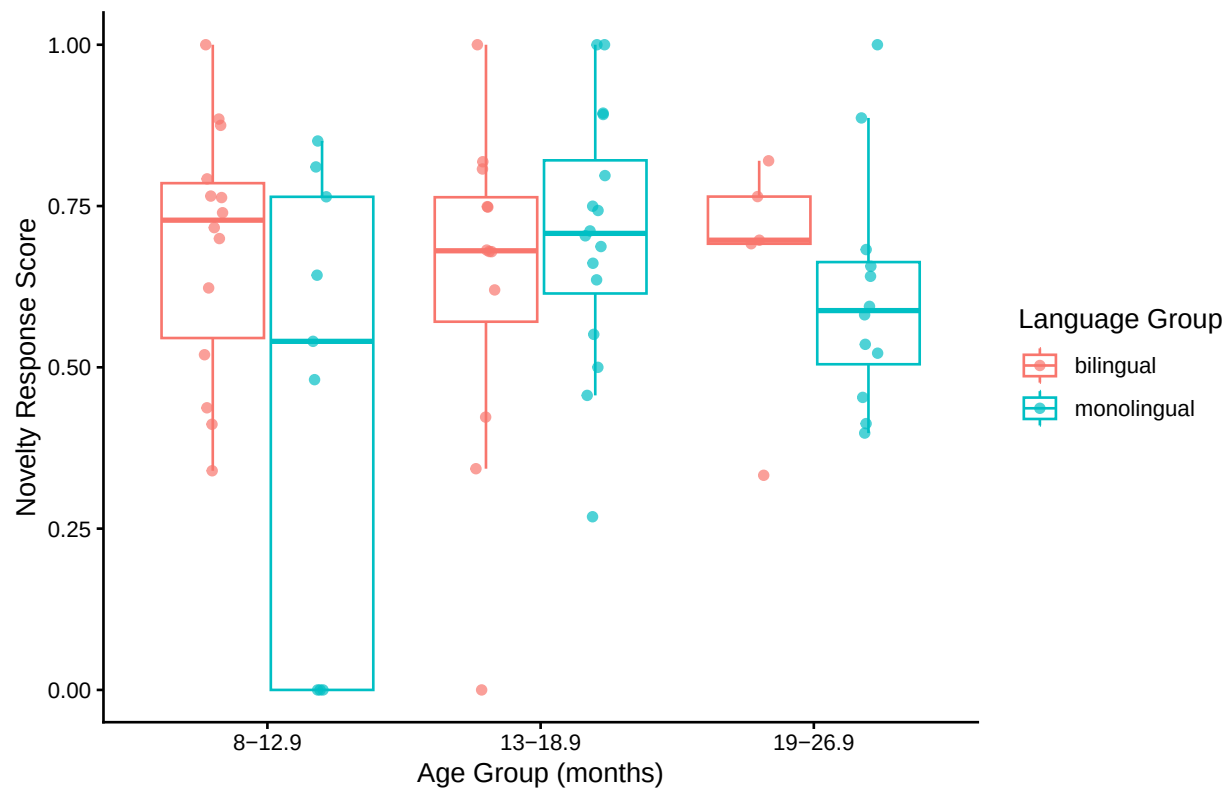
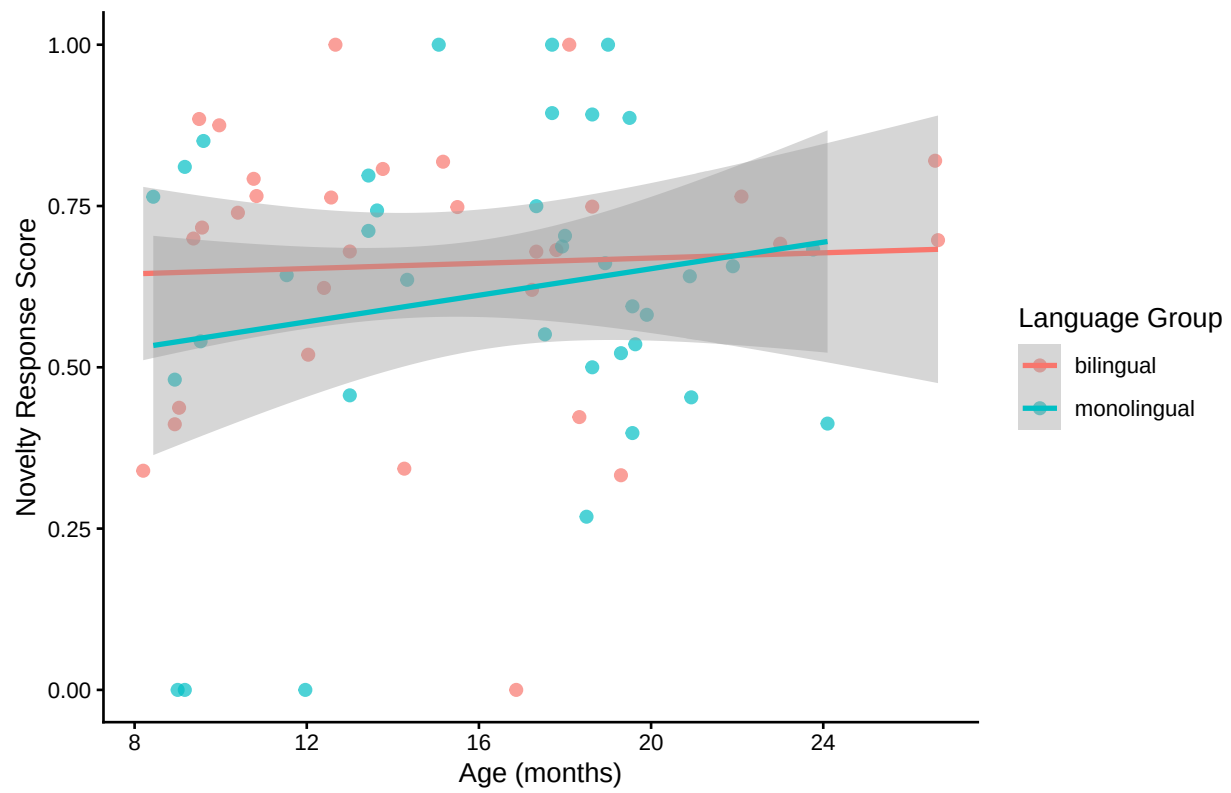
	r.squared	adj.r.squared	sigma	statistic	p.value	df	logLik	AIC	BIC	deviance
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	0.138	0.0542	0.227	1.64	0.182	4	5.58	0.835	11.8	2.11

i 2 more variables: df.residual <int>, nobs <int>

N = 46

Novelty Response Plots





Habituation + LENA Plots

